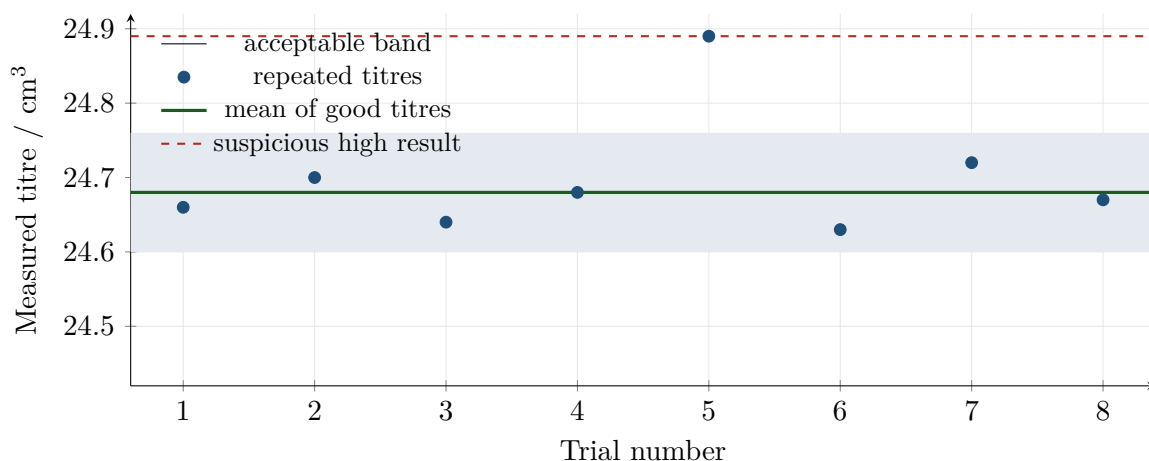


Statistical Methods in Chemistry

Block 1

Measurement, Error, and Uncertainty



Purpose of this block

By the end of this block, students should be able to report chemical measurements as estimates with uncertainty, identify random and systematic error, calculate percentage uncertainty, combine uncertainties in simple calculations, and interpret basic plots of repeated measurements.

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1 The central idea: chemical measurements are estimates

A chemical measurement is not the true value itself. It is an **estimate** produced by an instrument, a method, and a human procedure. In analytical chemistry, a result is scientifically weak if it is reported without any indication of uncertainty.

Core model of measurement

A useful way to think about any measurement is:

$$\text{measured value} = \text{true value} + \text{random error} + \text{systematic error}.$$

The true value is usually unknown. Statistics helps us estimate how reliable the measured value is.

In school chemistry, the same logic applies to familiar practical work:

- a titre from a burette;
- a mass from a balance;
- an absorbance reading from a colorimeter;
- a pH value from a pH meter;
- a time measurement during a rate experiment;
- a concentration calculated from a calibration curve or titration.

Why this matters

Two students may both calculate $0.100 \text{ mol dm}^{-3}$ for an acid concentration. But one student may have used careful repeated titres and a calibrated pipette, while another may have used one rough titre and an unstandardized base. The numerical answer alone hides the quality of the evidence.

Quick exercise: Measurement as evidence

For each result below, write whether it is strong or weak scientific reporting. Give one reason.

- "The concentration was $0.105 \text{ mol dm}^{-3}$."
- "The concentration was $(0.105 \pm 0.003) \text{ mol dm}^{-3}$ based on four concordant titres."
- "The unknown had absorbance 0.42, so its concentration was 0.20 mol dm^{-3} . No calibration graph is shown."

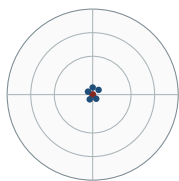
2 Accuracy and precision

Definitions

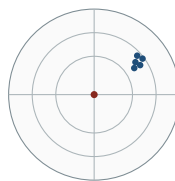
Accuracy is closeness to the true or accepted value.

Precision is closeness of repeated measurements to each other.

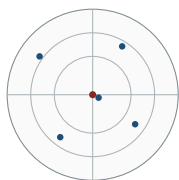
A set of measurements can be precise but inaccurate if they cluster tightly around the wrong value.



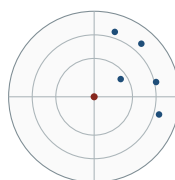
Accurate and precise
close to target; tightly grouped



Precise but inaccurate
tightly grouped; biased



Accurate on average
scattered around target



Neither accurate nor precise
scattered and biased

Quick exercise: Accuracy versus precision

A student measures the concentration of a standard hydrochloric acid solution. The accepted value is $0.1000 \text{ mol dm}^{-3}$.

Student	Trial 1	Trial 2	Trial 3	Trial 4
A	0.0998	0.1001	0.1000	0.0999
B	0.1120	0.1122	0.1119	0.1121
C	0.0940	0.1025	0.1080	0.0965
D	0.0840	0.1180	0.0910	0.1300

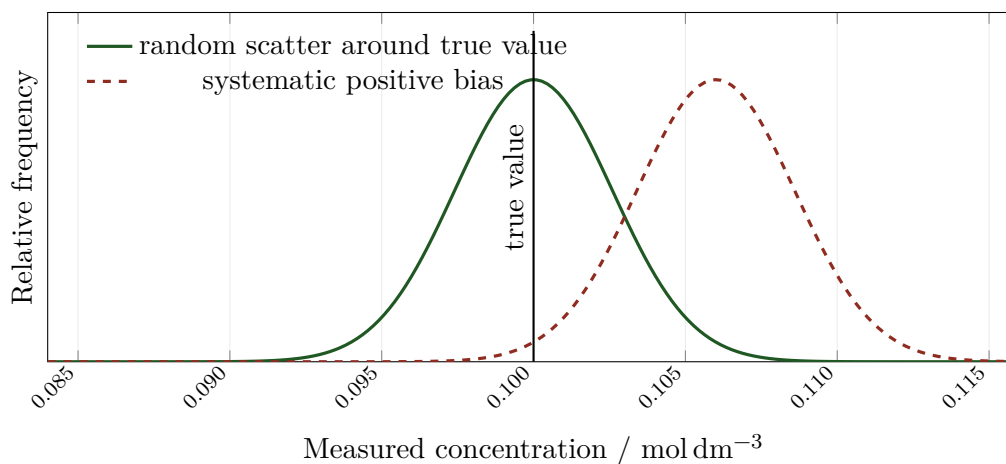
Classify each student's results as accurate, precise, both, or neither.

3 Random error and systematic error

Two major error types

Random error causes scatter in repeated measurements. Averaging repeated measurements can reduce its effect on the final mean.

Systematic error shifts measurements in one direction. Repetition alone does not remove it.



Common chemistry examples

Situation	Random error	Systematic error
Titration	Slight variation in judging the endpoint colour	NaOH standard solution absorbs CO ₂ and becomes less concentrated than assumed
Mass measurement	Balance reading fluctuates slightly	Balance not zeroed before use
Colorimetry	Small noise in absorbance reading	Blank was not used or cuvette has fingerprints
pH measurement	Reading drifts during stabilization	pH probe not calibrated
Rate experiment	Stopwatch reaction time varies	Reaction temperature is consistently higher than recorded

Important distinction

Repeating an experiment helps with random error. It does not automatically fix systematic error. A wrong standard solution gives precise wrong answers.

Quick exercise: Classifying errors

Classify each as random error or systematic error.

- a) A student reads the meniscus slightly differently each time.
 - b) A balance is not tared before every mass measurement.
 - c) A colorimeter cuvette has a fingerprint on the clear side for every reading.
 - d) The final drop in a titration sometimes overshoots the endpoint.
 - e) A pH probe is calibrated using an expired buffer solution.
-
-

4 Instrumental uncertainty

Every instrument has limited resolution. This gives a minimum uncertainty even when technique is good.

Absolute and percentage uncertainty

The **absolute uncertainty** has the same unit as the measurement, for example $\pm 0.05 \text{ cm}^3$.
The **percentage uncertainty** compares the uncertainty with the size of the measurement:

$$\% \text{ uncertainty} = \frac{\text{absolute uncertainty}}{\text{measured value}} \times 100\%.$$

Instrument	Typical reading uncertainty	Notes
Digital balance	± 0.001 g or ± 0.01 g	depends on balance resolution
Volumetric pipette	printed tolerance, e.g. ± 0.03 cm ³	usually more accurate than measuring cylinder
Volumetric flask	printed tolerance, e.g. ± 0.2 cm ³ for 250 cm ³	used for standard solutions
Burette reading	often ± 0.05 cm ³ per reading	a titre uses two readings, so combine them
Thermometer	often ± 0.5 °C or ± 0.1 °C	depends on scale/resolution
Stopwatch	commonly ± 0.01 s, but human reaction time is larger	practical uncertainty may exceed display resolution

Worked example: Burette titre uncertainty

A burette is read as 0.15 cm³ initially and 24.80 cm³ finally. Each reading has uncertainty ± 0.05 cm³.

$$\text{titre} = 24.80 - 0.15 = 24.65 \text{ cm}^3$$

Because the titre depends on two readings:

$$\text{absolute uncertainty} = 0.05 \text{ cm}^3 + 0.05 \text{ cm}^3 = 0.10 \text{ cm}^3.$$

$$\% \text{ uncertainty} = \frac{0.10}{24.65} \times 100 = 0.41\%.$$

So the titre should be reported as:

$$(24.65 \pm 0.10) \text{ cm}^3.$$

Quick exercise: Instrumental uncertainty

A student measures 12.40 cm³ of solution using a burette. Assume the titre uncertainty is ± 0.10 cm³.

- Calculate the percentage uncertainty.
- Explain why small titres have larger percentage uncertainty than large titres.

5 Significant figures and reporting results

Significant figures are not a decoration. They communicate the precision justified by the measurement.

Practical reporting rule

Usually report the uncertainty to **one significant figure**, or two if the first digit is 1 or 2. Then round the measured value to the same decimal place as the uncertainty.

Unrounded calculation	Better report	Reason
0.104738 ± 0.002914	0.1047 ± 0.0029 or 0.105 ± 0.003	uncertainty controls useful precision
24.653 ± 0.100	24.65 ± 0.10	same decimal place
1.2837 ± 0.0831	1.28 ± 0.08	do not overstate precision

Quick exercise: Rounding with uncertainty

Round each result appropriately.

a) $0.098763 \pm 0.004812 \text{ mol dm}^{-3}$

b) $18.436 \pm 0.214 \text{ s}$

c) $2.5038 \pm 0.0315 \text{ g}$

6 Combining uncertainties in calculations

Chemical results often require calculations: concentration from mass and volume, rate from volume and time, or amount of substance from mass and molar mass. The uncertainty in the final answer depends on the uncertainties of the input measurements.

Basic uncertainty propagation rules

For **addition or subtraction**, add absolute uncertainties:

$$q = a + b \quad \text{or} \quad q = a - b, \quad \Delta q = \Delta a + \Delta b.$$

For **multiplication or division**, add percentage uncertainties:

$$q = \frac{ab}{c}, \quad \% \Delta q = \% \Delta a + \% \Delta b + \% \Delta c.$$

For a **power**, multiply the percentage uncertainty by the power:

$$q = a^n, \quad \% \Delta q = n(\% \Delta a).$$

Worked example: Standard solution from mass and volume

A student prepares 250.0 cm^3 of Na_2CO_3 solution using 1.325 g of solid. The balance uncertainty is $\pm 0.001 \text{ g}$. The volumetric flask uncertainty is $\pm 0.2 \text{ cm}^3$. Assume $M_r(\text{Na}_2\text{CO}_3) = 106.0$.

$$n = \frac{m}{M_r} = \frac{1.325}{106.0} = 0.01250 \text{ mol}$$

$$c = \frac{n}{V} = \frac{0.01250}{0.2500} = 0.05000 \text{ mol dm}^{-3}$$

Percentage uncertainties:

$$\% \Delta m = \frac{0.001}{1.325} \times 100 = 0.075\%$$

$$\% \Delta V = \frac{0.2}{250.0} \times 100 = 0.080\%$$

For concentration, mass and volume are combined by division, so add percentage uncertainties:

$$\% \Delta c = 0.075\% + 0.080\% = 0.155\%.$$

Absolute uncertainty in concentration:

$$\Delta c = 0.00155 \times 0.05000 = 0.0000775 \text{ mol dm}^{-3}.$$

A sensible report is:

$$c = (0.05000 \pm 0.00008) \text{ mol dm}^{-3}.$$

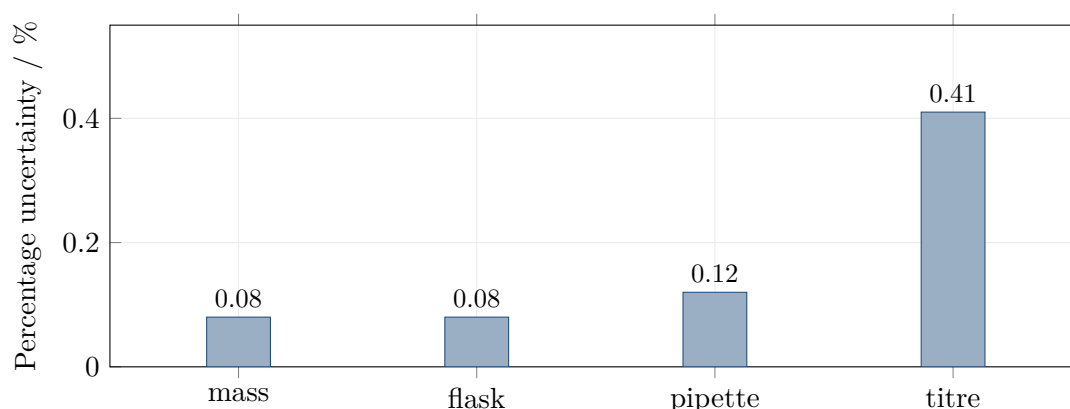
Quick exercise: Preparing a standard solution

A student dissolves 2.650 g of Na_2CO_3 and makes the solution up to 500.0 cm^3 . Use $M_r(\text{Na}_2\text{CO}_3) = 106.0$. The balance uncertainty is $\pm 0.001\text{ g}$. The volumetric flask uncertainty is $\pm 0.3\text{ cm}^3$.

- Calculate the concentration.
- Calculate the percentage uncertainty from the mass.
- Calculate the percentage uncertainty from the volume.
- Calculate the total percentage uncertainty in concentration.
- Report the concentration with uncertainty.

Which measurement dominates the uncertainty?

The plot below shows example percentage uncertainty contributions in a simple titration calculation. The largest bar shows the measurement that most limits the final precision.

**Quick exercise: Reading an uncertainty plot**

Use the bar chart above.

- Which measurement contributes the most uncertainty?
- Which improvement would reduce the final uncertainty more: using a better balance or obtaining a larger titre? Explain.

7 Repeated measurements: mean and spread

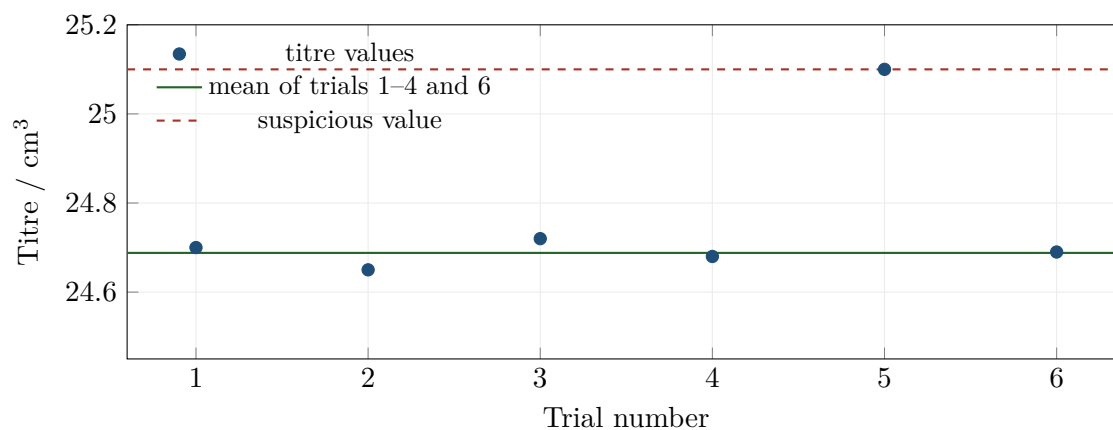
This block is mainly about uncertainty from instruments and calculations. However, repeated measurements already introduce a statistical idea: the **mean** is the best central estimate when random errors are present.

Mean of repeated values

For values $x_1, x_2, \dots, x_n$, the mean is:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

A mean should not be calculated blindly. First ask whether any value has a justified experimental reason for exclusion.



Do not delete data just because it is inconvenient

A value can be excluded only if there is a defensible reason: for example, the endpoint was overshoot, the burette leaked, a bubble was seen in the jet, or the cuvette was visibly contaminated. Excluding data only because it improves the conclusion is bad scientific practice.

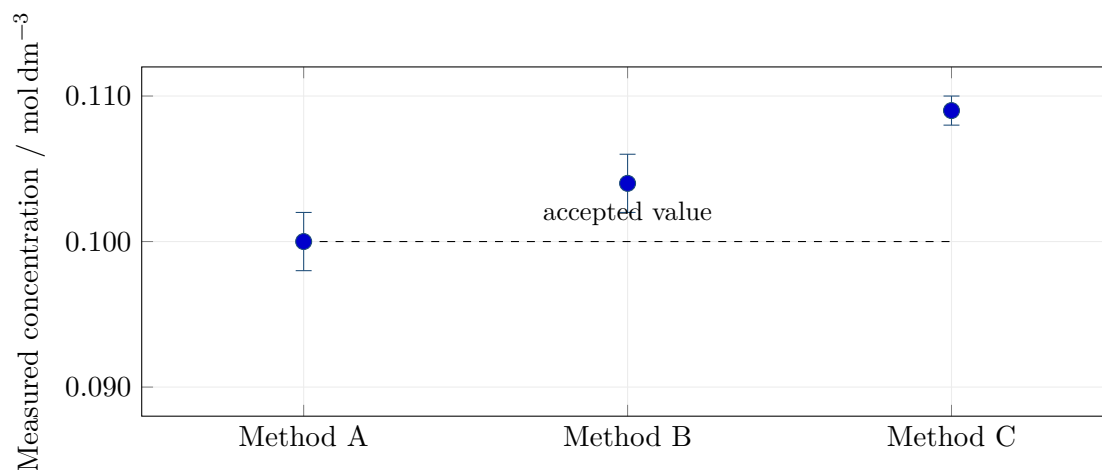
Quick exercise: Mean titre and suspicious value

Use the titre plot above.

- Calculate the mean of all six trials.
- Calculate the mean after excluding trial 5.
- What experimental reason would justify excluding trial 5?
- If no experimental reason was recorded, should trial 5 be excluded? Explain.

8 Graphing uncertainty: error bars and agreement

Error bars show uncertainty visually. Two measurements may look different, but the difference may not be meaningful if their uncertainty intervals overlap.



Quick exercise: Interpreting error bars

Use the error-bar plot above.

- Which method is closest to the accepted value?
- Which methods have uncertainty intervals that include 0.100 mol dm⁻³?
- Which method appears clearly biased high?
- Why is it dangerous to compare only the central dots and ignore the error bars?

9 Mini-practical data task

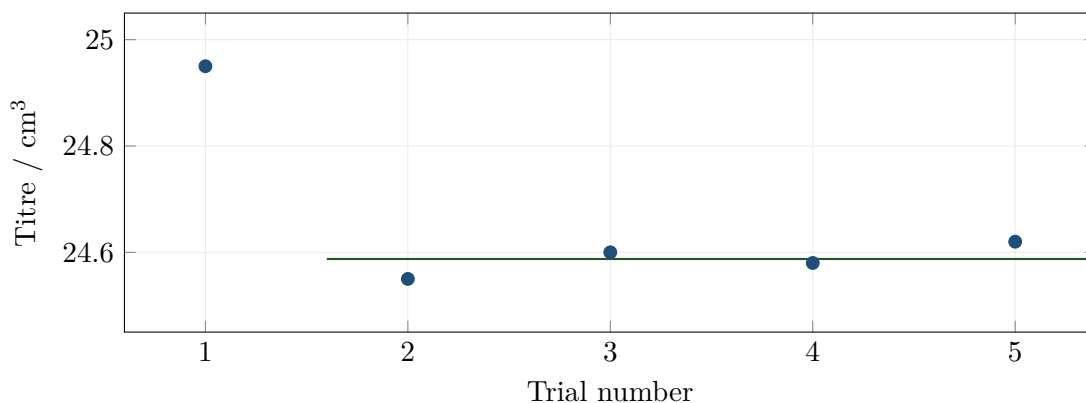
This task can be completed in class without a laboratory. Students work from the data table, calculate uncertainty, and interpret the plot.

Quick exercise: Acid-base titration data

A student standardizes hydrochloric acid using sodium carbonate solution. The burette uncertainty for each titre is $\pm 0.10 \text{ cm}^3$. The following titres were obtained:

Trial	1	2	3	4	5
Titre / cm ³	24.95	24.55	24.60	24.58	24.62

- Which trial is probably the rough titre?
- Calculate the mean titre using suitable trials.
- Calculate the percentage uncertainty of the mean titre using $\pm 0.10 \text{ cm}^3$ as the titre uncertainty.
- Report the mean titre with absolute uncertainty.
- Explain one possible systematic error in this titration.



10 Checklist for reporting a measurement

Before reporting a chemical result, check the following.

Question	What to look for
What was measured directly?	mass, volume, time, pH, absorbance, temperature
What instrument was used?	balance, burette, pipette, volumetric flask, colorimeter, pH meter
What is the instrument uncertainty?	absolute uncertainty with units
Was the measurement repeated?	calculate a mean only from justified values
Is random error visible?	scatter in repeated trials
Could systematic error be present?	calibration, contamination, wrong standard, heat loss, endpoint bias
How should the final value be rounded?	match decimal places between result and uncertainty
Does the graph support the conclusion?	error bars, trend, suspicious points, overlap

Quick exercise: Exit ticket

Answer in two or three sentences.

A student says: "My result is precise because my calculated concentration has many decimal places." Explain why this statement is wrong.

Answer key

Measurement as evidence. a) Weak: no uncertainty or repeats. b) Stronger: uncertainty and repeated concordant titres are stated. c) Weak: no calibration evidence and no uncertainty.

Accuracy versus precision. A: accurate and precise. B: precise but inaccurate. C: roughly accurate on average but not precise. D: neither accurate nor precise.

Classifying errors. a) Random. b) Systematic. c) Systematic. d) Random, unless the same student consistently overshoots every endpoint, in which case it becomes a systematic technique bias. e) Systematic.

Instrumental uncertainty. a) $(0.10/12.40) \times 100 = 0.806\% \approx 0.81\%$. b) The same absolute uncertainty is divided by a smaller measured value, so the percentage becomes larger.

Rounding with uncertainty. a) approximately $0.099 \pm 0.005 \text{ mol dm}^{-3}$. b) approximately $18.44 \pm 0.21 \text{ s}$, or $18.4 \pm 0.2 \text{ s}$ if using one significant figure in the uncertainty. c) approximately $2.504 \pm 0.032 \text{ g}$, or $2.50 \pm 0.03 \text{ g}$.

Preparing a standard solution.

$$n = \frac{2.650}{106.0} = 0.02500 \text{ mol}, \quad c = \frac{0.02500}{0.5000} = 0.05000 \text{ mol dm}^{-3}.$$

$$\% \Delta m = (0.001/2.650) \times 100 = 0.0377\%.$$

$$\% \Delta V = (0.3/500.0) \times 100 = 0.0600\%.$$

Total percentage uncertainty = $0.0977\% \approx 0.10\%$. Absolute uncertainty = $0.0000489 \text{ mol dm}^{-3}$. Report approximately $0.05000 \pm 0.00005 \text{ mol dm}^{-3}$.

Reading an uncertainty plot. a) The titre contributes most. b) Obtaining a larger titre, or otherwise reducing titre uncertainty, would reduce final uncertainty more than improving the already-small balance contribution.

Mean titre and suspicious value. Mean of all six trials:

$$\frac{24.70 + 24.65 + 24.72 + 24.68 + 25.10 + 24.69}{6} = 24.7567 \text{ cm}^3.$$

Mean excluding trial 5:

$$\frac{24.70 + 24.65 + 24.72 + 24.68 + 24.69}{5} = 24.688 \text{ cm}^3.$$

A valid reason might be overshooting the endpoint, leaking burette, or air bubble in the jet. Without a recorded reason, exclusion is questionable.

Interpreting error bars. a) Method A. b) Method A includes 0.100; Method B has interval approximately 0.102 to 0.106, so it does not include 0.100; Method C does not. c) Method C, clearly. d) Central dots ignore uncertainty; apparent differences may not be meaningful if uncertainty intervals overlap or include the accepted value.

Acid-base titration data. Trial 1 is probably rough. Suitable mean from trials 2–5:

$$\frac{24.55 + 24.60 + 24.58 + 24.62}{4} = 24.5875 \text{ cm}^3 \approx 24.59 \text{ cm}^3.$$

Percentage uncertainty $(0.10/24.59) \times 100 = 0.407\% \approx 0.41\%$. Report $24.59 \pm 0.10 \text{ cm}^3$. Possible systematic errors: wrong standard concentration, uncalibrated pipette, endpoint indicator bias, NaOH absorbing CO_2 , contaminated glassware.

Exit ticket. Decimal places from a calculator do not prove precision. Precision depends on the measurement process, instrument uncertainty, and scatter of repeated values. A result should be rounded according to the uncertainty.